CBSE Class 10 Mathematics Standard Question Paper 2024

Set - 1 (30/2/1) Solutions

SECTION – A

This section consists of 20 questions of 1 mark each.

1. The value of k for which the system of equations $3x-y+8=0$ and $6x-\text{ky}+16=0$ has infinitely many solutions, is

(A) -2

(B) 2

(C) $\frac{1}{2}$

(D) $-\frac{1}{2}$

Ans: (B) 2

2. Point P divides the line segment joining the points A(4,-5) and B(1,2) in the ratio 5:2. Co-ordinates of point P are

(A) $\left(\frac{5}{2},\frac{-3}{2}\right)$

(B) $\left(\frac{11}{7},0\right)$

(C) $\left(\frac{13}{7},0\right)$

(D) $\left(0,\frac{13}{7}\right)$

Ans: (C) $\left(\frac{13}{7},0\right)$

3. The common difference of an A.P. in which $a_{15}-a_{11}=48$**, is**

(A) 12

(B) 16

(C) -12

(D) -16

Ans: (A) 12

4. The quadratic equation $x^2+x+1=0$ **has ___ roots.**

(A) real and equal

(B) irrational

(C) real and distinct

(D) not-real

Ans: (D) not-real

5. If the $\text{HCF}(2520, 6600) = 40$ **and** $\text{LCM}(2520, 6600) = 252 \times k$**, then the value of k is**

(A) 1650

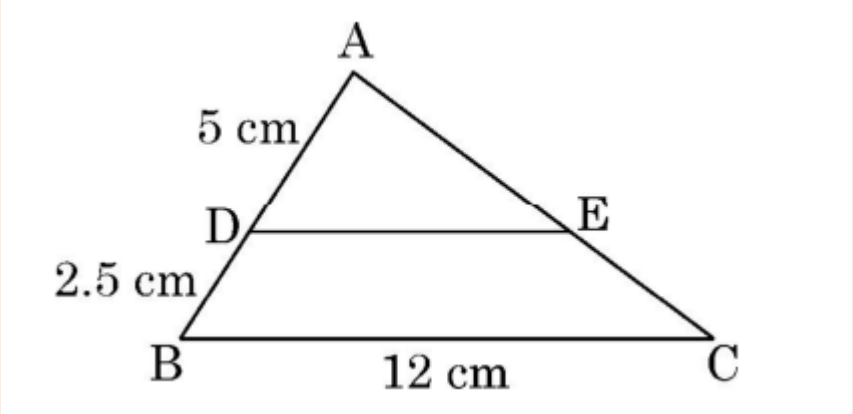
(B) 1600

(C) 165

(D) 1625

Ans: (C) 165

6. In the given figure $\triangle ABC$ **is shown. DE is parallel to BC . If** $AD = 5\text{cm}$, $DB = 2.5\text{cm}$ **and** $BC = 12\text{cm}$**, then DE is equal to**



(A) 10 cm

(B) 6 cm

(C) 8 cm

(D) 7.5 cm

Ans: (C) 8 cm

7. If $\sin\theta = \cos\theta, \left(0^\circ < \theta < 90^\circ\right)$**, then value of** $(\sec\theta \cdot \sin\theta)$ **is :**

(A) $\frac{1}{\sqrt{2}}$

(B) $\sqrt{2}$

(C) 1

(D) 0

Ans: (C) 1

8. Two dice are rolled together. The probability of getting the sum of the two numbers to be more than 10 , is

(A) $\frac{1}{9}$

(B) $\frac{1}{6}$

(C) $\frac{7}{12}$

(D) $\frac{1}{12}$

Ans: (D) $\frac{1}{12}$

9. If α and β are zeroes of the polynomial $5x^2+3x-7$, the value of $\frac{1}{\alpha}+\frac{1}{\beta}$ is

(A) $-\frac{3}{7}$

(B) $\frac{3}{5}$

(C) $\frac{3}{7}$

(D) $-\frac{5}{7}$

Ans: (D) $-\frac{5}{7}$

10. The perimeters of two similar triangles ABC and PQR are 56 cm and 48 cm respectively. PQ/AB **is equal to**

(A) $\frac{7}{8}$

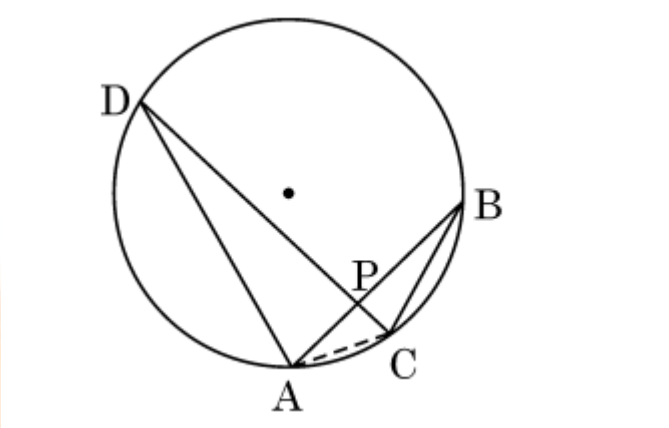
(B) $\frac{6}{7}$

(C) $\frac{7}{6}$

(D) $\frac{8}{7}$

Ans: (B) $\frac{6}{7}$

11. AB and CD are two chords of a circle intersecting at P . Choose the correct statement from the following:



(A) $\triangle ADP \sim \triangle CBA$

(B) $\triangle ADP \sim \triangle BPC$

(C) $\triangle ADP \sim \triangle BCP$

(D) $\triangle ADP \sim \triangle CBP$

Ans: (B) $\triangle ADP \sim \triangle BPC$

12. If value of each observation in a data is increased by 2 , then median of the new data

(A) increases by 2

(B) increases by 2 n

(C) remains same

(D) decreases by 2

Ans: (A) increases by 2

13. A box contains cards numbered 6 to 55 . A card is drawn at random from the box. The probability that the drawn card has a number which is a perfect square, is

(A) $\frac{7}{50}$

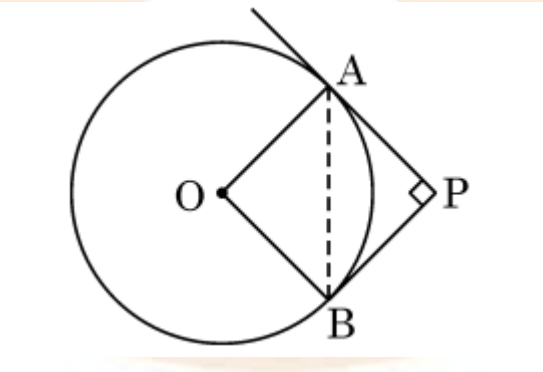
(B) $\frac{7}{55}$

(C) $\frac{1}{10}$

(D) $\frac{5}{49}$

Ans: (C) $\frac{1}{10}$

14. In the given figure, tangents PA and PB to the circle centred at O , from point P are perpendicular to each other. If $PA = 5\text{cm}$**, then length of A B is equal to**



(A) 5 cm

(B) $5\sqrt{2}\,\text{cm}$

(C) $2\sqrt{5}\,\text{cm}$

(D) 10 cm

Ans: (B) $5\sqrt{2}\,\text{cm}$

15. XOYZ is a rectangle with vertices $X(-3,0), O(0,0), Y(0,4)$ **and** $Z(x,y)$**. The length of its each diagonal is**

(A) 5 units

(B) $\sqrt{5}$ **units**

(C) x^2+y^2 **units**

(D) 4 units

Ans: (A) 5 units

16. Which term of the A.P. $-29,-26,-23,\ldots..61$ **is 16?**

(A) 11th

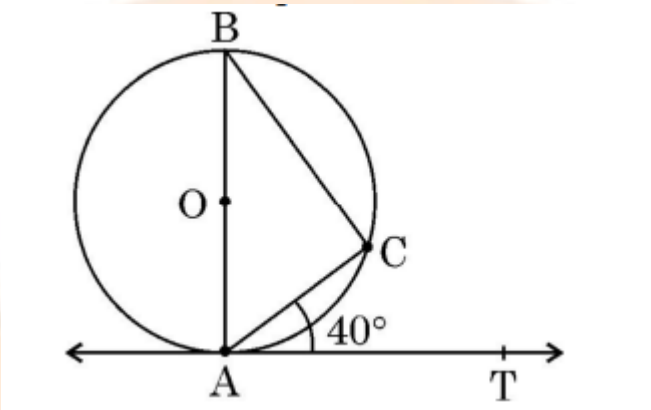
(B) 16th

(C) 10th

(D) 31^{*t}

Ans: (B) 16th

17. In the given figure, AT is tangent to a circle centred at O . If $\angle CAT = 40^\circ$, then $\angle CBA$ is equal to



(A) 70°

(B) 50°

(C) 65°

(D) 40°

Ans: (B) 50°

18. After an examination, a teacher wants to know the marks obtained by maximum number of the students in her class. She requires to calculate ___ of marks.

(A) median

(B) mode

(C) mean

(D) range

Ans: (B) mode

Directions : In Question 19 and 20, Assertion (A) and Reason (R) are given. Select the correct option from the following :

(A) Both Assertion (A) and Reason (R) are true. Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true. Reason (R) does not give correct explanation of (A).

(C) Assertion (A) is true but Reason (R) is not true.

(D) Assertion (A) is not true but Reason (R) is true.

19. Assertion (A) : If $\sin A = \frac{1}{3}\left(0^\circ < A < 90^\circ\right)$**, then the value of** $\cos A$ **is** $\frac{2\sqrt{2}}{3}$

Reason (R) : For every angle $\theta, \sin^2\theta + \cos^2\theta = 1$**.**

Ans: (A) Both Assertion (A) and Reason (R) are true. Reason (R) is the correct explanation of Assertion (A).

20. Assertion (A) : Two cubes each of edge length 10 cm are joined together. The total surface area of newly formed cuboid is $1200\,\text{cm}^2$**.**

Reason (R): Area of each surface of a cube of side 10 cm is $100\,\text{cm}^2$**.**

Ans: (D) Assertion (A) is not true but Reason (R) is true.

SECTION - B

In this section, there are 5 questions of 2 marks each.

21. Can the number $(15)^n$, n being a natural number, end with the digit 0 ? Give reasons.

Ans: For a number to end with the digit 0, it must be divisible by 10. Therefore, it must have both 2 and 5 as prime factors.

The number $(15)^n$ can be expressed as $15^n = (3\times5)^n = 3^n \times 5^n$.

Since $(15)^n$ only contains the factors 3 and 5, but no factor of 2, it cannot be divisible by 10.

Final Answer: No, $(15)^n$ cannot end with the digit 0 because it is not divisible by 10 (it lacks a factor of 2).

22. Find the type of triangle ABC formed whose vertices are $A(1,0), B(-5,0)$ and $C(-2,5)$.

Ans:

Calculate the lengths of the sides AB, BC, and CA:

1. Length of AB:

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(-5-1)^2 + (0-0)^2} = \sqrt{(-6)^2} = 6$$

2. Length of BC:

$$BC = \sqrt{(-2-(-5))^2 + (5-0)^2} = \sqrt{(3)^2 + (5)^2} = \sqrt{9+25} = \sqrt{34}$$

3. Length of CA:

$CA = \sqrt{(1-(-2))^2 + (0-5)^2} = \sqrt{(3)^2 + (-5)^2} = \sqrt{9+25} = \sqrt{34}$

Since $BC = CA = \sqrt{34}$ and AB=6, the triangle ABC has two sides equal. Triangle ABC is an isosceles triangle.

23. (a) Evaluate: $2\sin^2 30^\circ \sec 60^\circ + \tan^2 60^\circ$**.**

Ans:

1. Evaluate $\sin 30^\circ, \sec 60^\circ$ and $\tan 60^\circ$

$\sin 30^\circ = \frac{1}{2}, \quad \sec 60^\circ = 2, \quad \tan 60^\circ = \sqrt{3}$

2. Substitute:

$2\sin^2 30^\circ \sec 60^\circ + \tan^2 60^\circ = 2 \times \left(\frac{1}{2}\right)^2 \times 2 + (\sqrt{3})^2$

$= 2 \times \frac{1}{4} \times 2 + 3 = 1 + 3 = 4$

The expression evaluates to **4**.

OR

(b) If $2\sin(A+B) = \sqrt{3}$ **and** $\cos(A-B) = 1$**, then find the measures of angles A and B.** $0 \le \mathrm{A}, \mathrm{B}, (\mathrm{A}+\mathrm{B}) \le 90^\circ$**.**

Ans:

1. Solve $2\sin(A+B) = \sqrt{3}$:

$\sin(A+B)=\frac{\sqrt{3}}{2}$

$\sin\theta=\frac{\sqrt{3}}{2}$ when $\theta=60^\circ$.

Therefore, $A+B=60^\circ$

2. Solve $\cos(A-B)=1$:

$\cos(A-B)=1$, $\cos\theta=1$ when $\theta=0^\circ$

Therefore, $A-B=0^\circ$ so $A=B$.

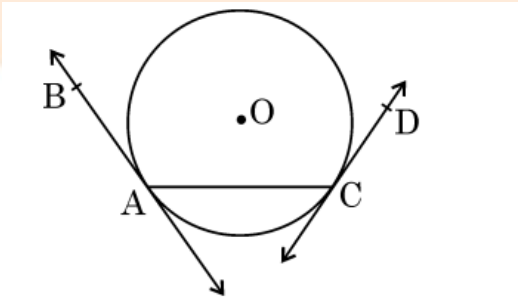
3. Substitute $A=B$ into $A+B=60^\circ$:

$2A=60^\circ \quad \Rightarrow \quad A=30^\circ$

Since A=B:
$B=30^\circ$

Final answer: $A=30^\circ, B=30^\circ$

24. In the given figure, AB and CD are tangents to a circle centred at O . Is $\angle BAC=\angle DCA$? Justify your answer.



Ans:

1. **Tangents from a common external point:**
   - AB and AO are radii, and AB is tangent to the circle at A, so $\angle OAB = 90^\circ$.
   - Similarly, CD and CO are radii, and CD is tangent to the circle at C, so $\angle OCD = 90^\circ$
2. **Angle Relationships:**
   - ∠BAC is the angle formed between the tangent AB and the chord AC.
   - ∠DCA is the angle formed between the tangent CD and the chord AC.
3. **Using the Tangent-Chord Theorem:**
   - The Tangent-Chord Theorem states that the angle between a tangent and a chord through the point of contact is equal to the angle in the alternate segment.
   - Therefore, ∠BAC=∠OCA (angle subtended by the chord AC in the opposite segment).
   - Similarly, ∠DCA=∠OAC.
4. **Comparing the Angles:**
   - ∠BAC=∠OCA and ∠DCA=∠OAC.
   - In a circle, the angles subtended by the same chord in the same segment are equal.

Therefore, ∠BAC=∠DCA.

25. (a) In what ratio is the line segment joining the points $(3,-5)$ and $(-1,6)$ divided by the line $y = x$?

Ans:

Let the line y=x divide the line segment joining the points (3,−5) and (−1,6) at point PPP. Since P lies on the line y=x, the coordinates of P will be (k,k).

Let's assume the ratio in which the line divides the segment is m:n.

Using the section formula, the coordinates of point P are given by:

$$P = \left(\frac{m(-1)+n(3)}{m+n}, \frac{m(6)+n(-5)}{m+n} \right)$$

Since P lies on the line y=x:

$$\frac{m(-1)+n(3)}{m+n} = \frac{m(6)+n(-5)}{m+n}$$
$$-m+3n = 6m-5n$$
$$-m-6m = -5n-3n$$
$$\frac{m}{n} = \frac{8}{7}$$

The line y=x divides the line segment joining the points (3,−5) and (−1,6) in the ratio 8:7.

OR

(b) $A(3,0), B(6,4)$ **and** $C(-1,3)$ **are vertices of a triangle ABC. Find length of its median BE.**

Ans: The median BE is drawn from vertex B to the midpoint E of side AC.

Find the coordinates of the midpoint E.

The midpoint formula for the points A(3,0) and C(−1,3) is:

$$E=\left(\frac{3+(-1)}{2},\frac{0+3}{2}\right)=\left(\frac{2}{2},\frac{3}{2}\right)=\left(1,\frac{3}{2}\right)$$

Find the length of median BE.

The length of BE is given by the distance formula:

$$BE=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$BE=\sqrt{(6-1)^2+\left(4-\frac{3}{2}\right)^2}$$

$$BE=\sqrt{5^2+\left(\frac{8-3}{2}\right)^2}=\sqrt{25+\left(\frac{5}{2}\right)^2}=\sqrt{25+\frac{25}{4}}$$

$$BE=\sqrt{\frac{100}{4}+\frac{25}{4}}=\sqrt{\frac{125}{4}}=\frac{\sqrt{125}}{2}=\frac{5\sqrt{5}}{2}$$

The length of the median BE is $\frac{5\sqrt{5}}{2}$ units.

SECTION - C

This section consists of 6 questions of 3 marks each.

26. (a) If the sum of first m terms of an A.P. is same as sum of its first n terms $(m\neq n)$, then show that the sum of its first (m+n) terms is zero.

Ans: Let the first term of the A.P. be a and the common difference be d.

The sum of the first mmm terms is given by:

$$S_m=\frac{m}{2}[2a+(m-1)d]$$

The sum of the first n terms is given by:

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

Given that $S_m = S_n$:

$$\frac{m}{2}[2a + (m-1)d] = \frac{n}{2}[2a + (n-1)d]$$
$$m[2a + (m-1)d] = n[2a + (n-1)d]$$
$$2am + m(m-1)d = 2an + n(n-1)d$$
$$2a(m-n) + d[m(m-1) - n(n-1)] = 0$$

Since $m \neq n$, m − n is not zero, so we can divide by m−n:

$$2a + d\frac{m^2 - m - n^2 + n}{m-n} = 0$$

The expression in the bracket simplifies to:

$$(m-n)(m+n) = 2a + d(m+n-1)$$
$$S_{m+n} = \frac{m+n}{2}[2a + ((m+n)-1)d] = 0$$

So the sum of the first m+n terms is zero.

OR

(b) In an A.P., the sum of three consecutive terms is 24 and the sum of their squares is 194. Find the numbers.

Ans: Let the three consecutive terms be a−d, a, and a+d.

1. Sum of the three terms:

$$(a-d)+a+(a+d)=3a=24$$
$$a=8$$

Sum of the squares of the three terms:

$$(a-d)^2+a^2+(a+d)^2=194$$
$$(8-d)^2+8^2+(8+d)^2=194(\because \text{a}=8)$$

Expand and simplify:

$$64-16d+d^2+64+64+16d+d^2=194$$
$$3d^2+192=194$$
$$3d^2=2 \quad \Rightarrow \quad d^2=\frac{2}{3}$$
$$d=\pm\sqrt{\frac{2}{3}}=\pm\frac{\sqrt{6}}{3}$$

The three numbers are $8-\frac{\sqrt{6}}{3}, 8, \text{and} 8+\frac{\sqrt{6}}{3}$.

27. Prove that $\sqrt{5}$ is an irrational number.

Ans:

Let's assume that $\sqrt{5}$ is rational. This means it can be expressed as a ratio of two integers $\frac{p}{q}$, where p and q have no common factors (i.e., the fraction is in its lowest terms) and q≠0.

$$\sqrt{5}=\frac{p}{q}$$

Square both sides:

$$5 = \frac{p^2}{q^2}$$

Multiply both sides by q^2:

$$p^2 = 5q^2$$

This implies that p^2 is divisible by 5, so p must also be divisible by 5 (because if a prime number divides a square, it must divide the original number).

Let p=5k for some integer k. Substitute this into the equation:

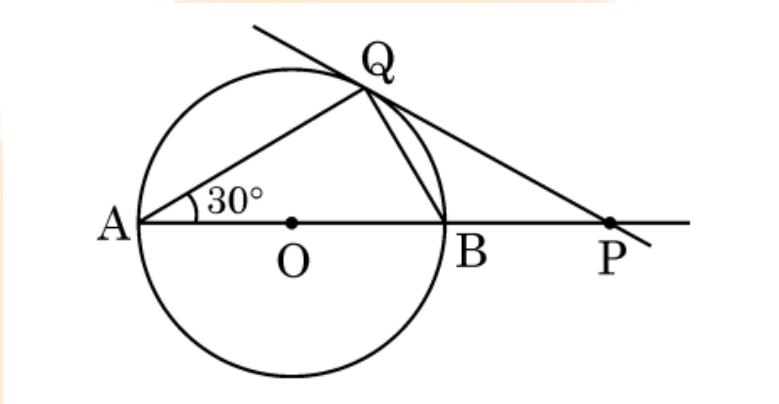
$$(5k)^2 = 5q^2$$
$$25k^2 = 5q^2$$
$$5k^2 = q^2$$

This implies that q^2 is divisible by 5, so q must also be divisible by 5.

But this means both p and q are divisible by 5, which contradicts the assumption that p and q have no common factors.

Therefore, $\sqrt{5}$ cannot be expressed as a ratio of two integers, so it is irrational.

28. (a) In the given figure, PQ is tangent to a circle centred at O and $\angle BAQ = 30^\circ$; show that $BP = BQ$.



Ans:

Given:

- PQ is tangent to a circle centred at O.
- $\angle BAQ = 30^\circ$

To Prove:

- BP=BQ.

Solution:

1. Since PQ is tangent at point Q, the radius OQ is perpendicular to PQ.
2. $\angle OQB = 90^\circ$.
3. In $\Delta OQB, \angle OQB = 90^\circ$ and $\angle BAQ = 30^\circ$, so:

$\angle OBQ = 90^\circ - 30^\circ = 60^\circ$

4. As ∠OBQ=∠OBA (alternate angles):

$\triangle OBQ \cong \triangle OBA$ (by SAS criterion)

5. Since the triangles are congruent, BP=BQ.

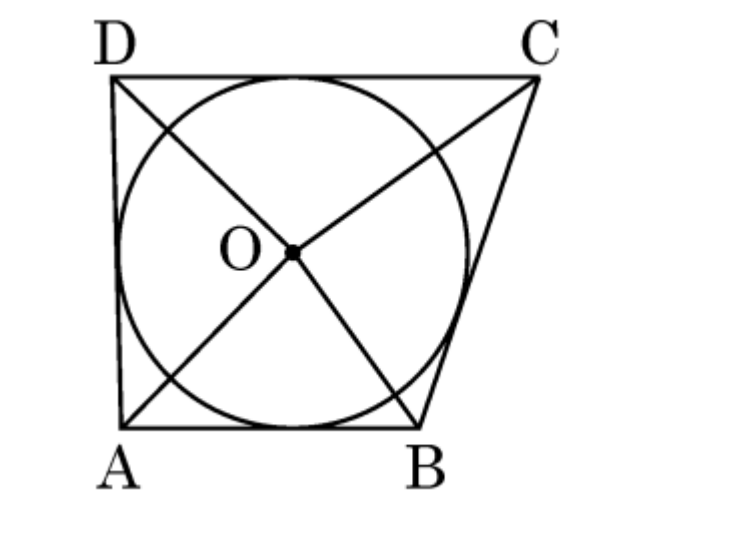
Final Answer:

- BP=BQ.

OR

(b) In the given figure, AB, BC, CD **and DA are tangents to the circle with centre O forming a quadrilateral A B C D .**

Show that $\angle AOB + \angle COD = 180^\circ$



Ans: Given:

- AB,BC,CD ,DA are tangents to the circle centred at O forming quadrilateral ABCD.

To Prove:

$\angle AOB + \angle COD = 180^\circ$

Solution:

1. AB and BC are tangents, and O is the centre. The angles subtended by these tangents at the centre are ∠AOB and ∠COD.

2. Since tangents from a point are equal in length and subtend equal angles at the centre, the sum of the angles subtended by opposite sides of the quadrilateral ABCD at the centre is:

$\angle AOB + \angle COD = 180^\circ$ (cyclic quadrilateral property)

Final Answer: $\angle AOB + \angle COD = 180^\circ$

29. Prove that $\frac{1+\sec\theta-\tan\theta}{1+\sec\theta+\tan\theta} = \frac{1-\sin\theta}{\cos\theta}$

Ans:

Given: $\frac{1+\sec\theta-\tan\theta}{1+\sec\theta+\tan\theta} = \frac{1-\sin\theta}{\cos\theta}$

Substitute $\sec\theta = \frac{1}{\cos\theta}$ and $\tan\theta = \frac{\sin\theta}{\cos\theta}$:

$$\frac{1+\frac{1}{\cos\theta}-\frac{\sin\theta}{\cos\theta}}{1+\frac{1}{\cos\theta}+\frac{\sin\theta}{\cos\theta}} = \frac{\frac{\cos\theta+1-\sin\theta}{\cos\theta}}{\frac{\cos\theta+1+\sin\theta}{\cos\theta}}$$

$$\frac{\cos\theta+1-\sin\theta}{\cos\theta+1+\sin\theta}$$

Multiply numerator and denominator by the conjugate of the denominator:

$$\frac{(\cos\theta+1-\sin\theta)^2}{(\cos\theta+1)^2-\sin^2\theta}$$

Simplify the denominator using the identity $(a^2-b^2=(a-b)(a+b))$:

$$(\cos\theta+1)^2-\sin^2\theta=1-\sin^2\theta+2\cos\theta+\cos^2\theta=2(1+\cos\theta)$$

$$\frac{(\cos\theta+1-\sin\theta)(\cos\theta+1-\sin\theta)}{2(1+\cos\theta)}$$

$$=\frac{(1-\sin\theta)(\cos\theta+1)}{2\cos\theta}$$

$$=\frac{1-\sin\theta}{\cos\theta}$$

Hence, $\frac{1+\sec\theta-\tan\theta}{1+\sec\theta+\tan\theta}=\frac{1-\sin\theta}{\cos\theta}$

30. In a test, the marks obtained by 100 students (out of 50) are given below :

Marks obtained :	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Number of students :	12	23	34	25	6

Find the mean marks of the students.

Ans:

To find the mean marks, we use the formula: Mean $=\frac{\sum f_i x_i}{\sum f_i}$

Where:

- f_i is the frequency (number of students).

- x_i is the midpoint of each class interval.

Class Intervals and Midpoints (x_i):

- 0–10: Midpoint $x_1 = 5$
- 10–20: Midpoint $x_2 = 15$
- 20–30: Midpoint $x_3 = 25$
- 30–40: Midpoint $x_4 = 35$
- 40–50: Midpoint $x_5 = 45$

2. Frequency f_i and Products $f_i x_i$:

- $f_1 = 12,\ 12 \times 5 = 60$
- $f_2 = 23, 23 \times 15 = 345$
- $f_3 = 34, 34 \times 25 = 850$
- $f_4 = 25,\ 25 \times 35 = 875$
- $f_5 = 6$, $6 \times 45 = 270$

3. Total Frequency:

$$\sum f_i = 12 + 23 + 34 + 25 + 6 = 100$$
$$\sum f_i x_i = 60 + 345 + 850 + 875 + 270 = 2400$$

Mean Calculation:

$$\text{Mean} = \frac{2400}{100} = 24$$

Mean marks of the students = 24.

31. In a 2 -digit number, the digit at the unit's place is 5 less than the digit at the ten's place. The product of the digits is 36 . Find the number.

Ans:

Step 1: Define the digits

- Let the ten's digit be x.
- Unit's digit = x−5.

Step 2: Set up the equation

- Product of the digits: $x \times (x-5) = 36$

Step 3: Solve the quadratic equation

$$x^2 - 5x - 36 = 0$$

$$x = \frac{5 \pm \sqrt{25+144}}{2} = \frac{5 \pm \sqrt{169}}{2} = \frac{5 \pm 13}{2}$$

Step 4: Find the value of x:

- x = 9 (since x = −4 is not possible).

Step 5: Find the unit's digit

- Unit's digit = 9 −5=4.

Final Answer:

The two-digit number is **94**.

SECTION - D

32. Tara scored 40 marks in a test, getting 3 marks for each right answer and losing 1 mark for each wrong answer. Had 4 marks been awarded for each correct answer and 2 marks been deducted for each wrong answer, then Tara would have scored 50 marks. Assuming that Tara attempted all questions, find the total number of questions in the test.

Ans:

First Scenario:

- Tara gets 3 marks for each correct answer.
- Tara loses 1 mark for each wrong answer.

The equation for the first scenario is:

$3x - y = 40$ (Equation 1)

Second Scenario:

- Tara gets 4 marks for each correct answer.
- Tara loses 2 marks for each wrong answer.

The equation for the second scenario is:

$4x - 2y = 50$ (Equation 2)

Solving the Equations:

1. From Equation 2, simplify:

Divide the entire equation by 2:

$2x - y = 25$ (Equation 3)

$$(3x-y)-(2x-y)=40-25$$
$$x=15$$

Substitute x=15 into Equation 3:

$$2(15)-y=25$$
$$30-y=25$$
$$y=5$$

Total Number of Questions:

$$x+y=15+5=20$$

The total number of questions in the test is 20.

33. (a) If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio.

Ans:

Given: A triangle ABC with a line DE drawn parallel to side BC, intersecting AB at D and AC at E.

To Prove:

$$\frac{AD}{DB}=\frac{AE}{EC}$$

Proof:

1. Consider Triangles △ADE and △ABC:

- Since DE∥BC, ∠ADE=∠ABC (corresponding angles are equal).
- Similarly, ∠DEA=∠BCA (corresponding angles are equal).

2. Write the Proportionality Condition from the Similarity:

   Since △ADE∼△ABC, the corresponding sides of the triangles are proportional:

   $$\frac{AD}{AB} = \frac{AE}{AC} \quad \text{and} \quad \frac{DE}{BC} = \frac{AD}{AB}$$

3. Express AB and AC in Terms of Segments:

We can write AB as AB=AD+DB and AC as AC=AE+EC.

Substituting these into the proportionality condition gives:

$$\frac{AD}{AD+DB} = \frac{AE}{AE+EC}$$
$$AD \cdot (AE+EC) = AE \cdot (AD+DB)$$
$$AD \cdot AE + AD \cdot EC = AE \cdot AD + AE \cdot DB$$

Since AD·AE appears on both sides, they cancel out, leaving:

$$AD \cdot EC = AE \cdot DB$$
$$\frac{AD}{DB} = \frac{AE}{EC}$$

This proves that if a line is drawn parallel to one side of a triangle, intersecting the other two sides, it divides those two sides in the same ratio.

OR

(b) Sides AB and AC and median AD to ΔABC are respectively proportional to sides PQ and PR and median PM of another triangle PQR. Show that $\Delta ABC \sim \Delta PQR$.

Ans: Given:

$$\frac{AB}{PQ} = \frac{AC}{PR} = \frac{AD}{PM}$$

To Prove: $\triangle ABC \sim \triangle PQR$

Steps:

1. Since $\frac{AB}{PQ} = \frac{AC}{PR}$, triangles $\triangle ABC$ and $\triangle PQR$ have their corresponding sides in proportion.
2. The given condition also states that the medians AD and PM are proportional.
3. By the Converse of the Basic Proportionality Theorem (or the Similarity Criterion):

ΔABC *is similar to* ΔPQR

Therefore, $\triangle ABC \sim \triangle PQR$.

34. From the top of a 45 m high light house, the angles of depression of two ships, on the opposite side of it, are observed to be 30° and 60°. If the line joining the ships passes through the foot of the light house, find the distance between the ships. (Use $\sqrt{3} = 1.73$)

Ans: Let the distances of the ships from the foot of the lighthouse be d_1 and d_2 on opposite sides of the lighthouse.

Using trigonometric ratios for each angle of depression:

1. For the ship at 30° :

$$\tan 30^\circ = \frac{\text{height of lighthouse}}{\text{distance from lighthouse}}$$

$$\frac{1}{\sqrt{3}} = \frac{45}{d_1}$$

$$d_1 = 45 \times \sqrt{3} = 45 \times 1.73 = 77.85 \text{ m}$$

2. For the ship at 60° :

$$\tan 60^\circ = \frac{\text{height of lighthouse}}{\text{distance from lighthouse}}$$

$$\sqrt{3} = \frac{45}{d_2}$$

$$d_2 = \frac{45}{\sqrt{3}} = \frac{45}{1.73} \approx 26.01 \text{ m}$$

Distance between the two ships:

$$\text{Distance} = d_1 + d_2 = 77.85 + 26.01 = 103.86 \text{ m}$$

The distance between the two ships is approximately 103.86 m.

35. The perimeter of a certain sector of a circle of radius 5.6 m is 20.0 m . Find the area of the sector.

Ans:

Given:

- Radius of the circle r=5.6 m
- Perimeter of the sector = 20.0 m

The perimeter of a sector is given where l is the length of the arc.

$$\text{Perimeter} = 2r + l$$
$$20.0 = 2(5.6) + l$$
$$l = 20.0 - 11.2 = 8.8 \text{ m}$$

The area of the sector is given by:

$$\text{Area} = \frac{1}{2} rl$$

Substitute r=5.6 m and l=8.8 m:

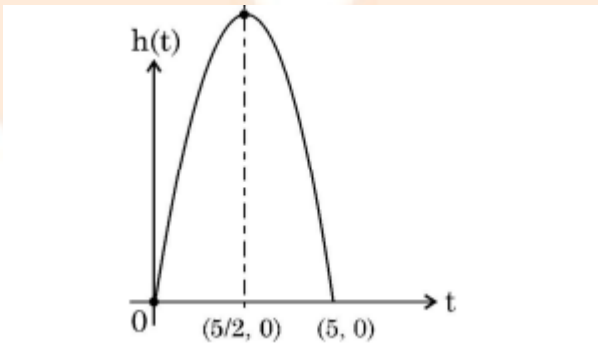
$$\text{Area} = \frac{1}{2} \times 5.6 \times 8.8 = 24.64 \text{ m}^2$$

The area of the sector is $\mathbf{24.64}\,\text{m}^2$.

SECTION - E

This section consists of 3 case-based questions of 4 marks each.

36. A ball is thrown in the air so that t seconds after it is thrown, its height h metre above its starting point is given by the polynomial $h = 25t - 5t^2$**.**



Observe the graph of the polynomial and answer the following questions :

(i) Write zeroes of the given polynomial.

Ans:

To find the zeroes, set h(t)=0:

$$25t - 5t^2 = 0$$
$$5t(5-t) = 0$$
$$t = 0 \quad \text{or} \quad t = 5$$

(ii) Find the maximum height achieved by ball.

Ans:

The maximum height occurs at the vertex of the parabola. The vertex occurs at:

$$t = \frac{-b}{2a} = \frac{-25}{2(-5)} = \frac{25}{10} = 2.5 \text{ seconds}$$

Substitute t=2.5 into the height equation:

$$h(2.5) = 25(2.5) - 5(2.5)^2 = 62.5 - 31.25 = 31.25 \text{ meters}$$

(iii) (a) After throwing upward, how much time did the ball take to reach to the height of 30 m?

Ans:

Set h(t)=30 and solve for t:

$$25t - 5t^2 = 30$$

Rearrange into standard quadratic form:

$$5t^2 - 25t + 30 = 0$$
$$t^2 - 5t + 6 = 0$$
$$(t-2)(t-3) = 0$$
$$t = 2 \text{ seconds} \quad \text{or} \quad t = 3 \text{ seconds}$$

The ball took 2 seconds to reach 30 m while going up.

OR

(iii) (b) Find the two different values of t when the height of the ball was 20 m .

Ans:

Set h(t)=20 and solve for t:

$$25t - 5t^2 = 20$$

Rearrange into standard quadratic form:

$$5t^2 - 25t + 20 = 0$$
$$t^2 - 5t + 4 = 0$$
$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{5 \pm \sqrt{25-16}}{2} = \frac{5 \pm 3}{2}$$
$$t = \frac{8}{2} = 4 \text{ seconds} \quad \text{or} \quad t = \frac{2}{2} = 1 \text{ second}$$

The ball was at a height of 20 meters at t=1 second and t=4 seconds.

37. The word 'circus' has the same root as 'circle'. In a closed circular area, various entertainment acts including human skill and animal training are presented before the crowd.

A circus tent is cylindrical upto a height of 8 m and conical above it. The diameter of the base is 28 m and total height of tent is 18.5 m .

Based on the above, answer the following questions:

(i) Find slant height of the conical part.

Ans:

- Radius of the base $r = \frac{28}{2} = 14$
- Height of the conical part h=18.5−8=10.5 m

The slant height l is given by:

$$l = \sqrt{r^2 + h^2} = \sqrt{14^2 + 10.5^2} = \sqrt{196 + 110.25} = \sqrt{306.25} \approx 17.5 \text{ m}$$

Final Answer: The slant height of the conical part is 17.5 m.

(ii) Determine the floor area of the tent.

Ans: Floor area is the area of the base (which is a circle).

Floor Area $= \pi r^2 = \pi \times 14^2 = \pi \times 196 \approx 615.44 \text{ m}^2$

The floor area of the tent is approximately 615.44 m².

(iii) (a) Find area of the cloth used for making tent.

Ans: Area of the cloth includes the lateral surface area of the cylinder and the lateral surface area of the cone.

1. Lateral Surface Area of Cylinder:

$A_{\text{cylinder}} = 2\pi rh = 2 \times \pi \times 14 \times 8 = 704\pi \text{ m}^2 \approx 2209.6 \text{ m}^2$

2. Lateral Surface Area of Cone:

$A_{\text{cone}} = \pi rl = \pi \times 14 \times 17.5 = 245\pi \text{ m}^2 \approx 769.3 \text{ m}^2$

3. Total Area of the Cloth:

$A_{\text{total}} = 2209.6 + 769.3 = 2978.9 \text{ m}^2$

The area of the cloth used is approximately 2978.9 m².

OR

(iii) (b) Find total volume of air inside an empty tent.

Ans:

- Volume of air inside the tent = Volume of the cylinder + Volume of the cone.

1. Volume of Cylinder:

$$V_{\text{cylinder}} = \pi r^2 h = \pi \times 14^2 \times 8 = 352\pi \text{ m}^3 \approx 1105.04 \text{ m}^3$$

2. Volume of Cone:

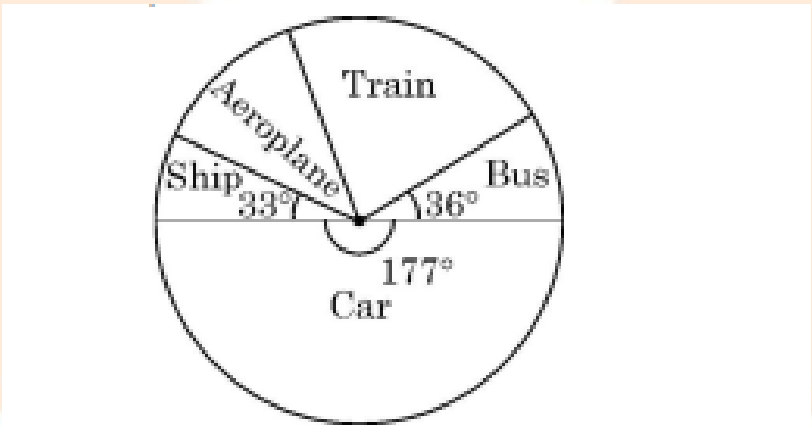
$$V_{\text{cone}} = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \pi \times 14^2 \times 10.5 = 686\pi \text{ m}^3 \approx 2154.24 \text{ m}^3$$

3. Total Volume:

$$V_{\text{total}} = 1105.04 + 2154.24 = 3259.28 \text{ m}^3$$

The total volume of air inside the empty tent is approximately 3259.28 m³.

38. In a survey on holidays, 120 people were asked to state which type of transport they used on their last holiday. The following pie chart shows the results of the survey.



Observe the pie chart and answer the following questions:

(i) If one person is selected at random, find the probability that he/she travelled by bus or ship.

Ans:

Calculate the combined angle for bus and ship:

$$\text{Bus} + \text{Ship} = 36^\circ + 24^\circ = 60^\circ$$

Probability:

$$P(\text{Bus or Ship}) = \frac{\text{Angle for Bus + Ship}}{360^\circ} = \frac{60^\circ}{360^\circ} = \frac{1}{6}$$

The probability is $\frac{1}{6}$.

(ii) Which is most favourite mode of transport and how many people used it?

Ans:

1. Most favorite mode:

- **Train** has the largest angle of 177°.

2. Number of people who used the train:

$$\text{Number of people} = \frac{177^\circ}{360^\circ} \times 120 = \frac{177 \times 120}{360} = 59 \text{ people}$$

The most favorite mode is **Train**, and 59 people used it.

(iii) (a) A person is selected at random. If the probability that he did not use train is $4/5$, find the number of people who used train.

Ans:

1. Probability of using the train:

$$P(\text{Train}) = 1 - \frac{4}{5} = \frac{1}{5}$$

2. Number of people who used the train:

$$\text{Number of people} = \frac{1}{5} \times 120 = 24 \text{ people}$$

24 people used the train.

OR

(iii) (b) The probability that randomly selected person used aeroplane is 7/60. Find the revenue collected by air company at the rate of ₹ 5,000 per person.

Ans:

1. Number of people who used aeroplane:

$$\text{Number of people} = \frac{7}{60} \times 120 = 14 \text{ people}$$

2. Revenue collected:

$$\text{Revenue} = 14 \times 5000 = \text{Rs.}70,000$$

The revenue collected by the air company is ₹ 70,000.